

Khinil Sahu

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Summary

MS by Research student in Civil Engineering at IIT Bhubaneswar with research focus in thermal barrier coatings analysis , corrosion-fatigue analysis of marine structures, fracture mechanics, and computational modeling. Completed a B.Tech in Civil Engineering from IGIT Sarang, including a final-year research project at IIT Bombay. Skilled in AutoCAD, Julia, Gmsh, and QGIS. Experienced in geotechnical investigations, material testing, technical writing, and research presentations. Strong believer in Parkinson's Law and the idea that anything can be learned through consistent effort and objective clarity.

Education

Indian Institute Of Technology (IIT) Bhubaneswar

July 2026 - Present

MS(R) in Civil Engineering

Research interests: Thermo-Chemo-Mechanical analysis , Corrosion-fatigue analysis on marine structures

Indira Gandhi Institute of Technology (IGIT), Sarang

Nov 2022 – Jun 2026

B.Tech in Civil Engineering

CGPA: 9.04

Work History

Project Research Intern – IIT Bombay

Feb 2026 – Jun 2026

- Studied fundamentals of **Ocean Engineering** and Tsunami Physics.
- Explored numerical approaches for **Tsunami modeling**.
- Created a **Tsunami hazard map for Tamil Nadu coastal regions** using QGIS and ADCIRC

Research Intern – IIT Bhubaneswar

May 2025 – Jul 2025

- Studied 5+ research papers on bolt loosening and presented technical findings.
- Studied bolt loosening under transverse load using **Finite Element Method (FEM)**.
- Applied **Gmsh** for meshing & **Julia + Gridap.jl** for FEM simulation and used **ParaView** for visualization.

Summer Intern – Govt. of Odisha (WR Department)

May 2024 – Jul 2024

- Assisted in site execution for structural and canal works.
- Supported estimation, billing, and field material testing.
- Observed infrastructure planning and project execution on-site.

Projects

Interface Shear Strength Between Fly Ash and Jute Geotextile

Aug 2025 – Dec 2025

- Conducted laboratory direct shear tests to study interaction between fly ash and jute geotextile.
- Determined shear strength parameters and evaluated geotechnical behavior.
- Proposed sustainable applications for ground improvement and slope stability.

Skills

- **Engineering Tools:** AutoCAD, Gmsh, ParaView ,
- **Programming & Analysis:** FEA, Ocean modeling, Python, Julia, MS Office (Excel, PowerPoint, Word), Overleaf
- **Domain Knowledge:** Steel & RCC Design, Surveying, Fluid & OCF, Geotechnical Engineering.
- **Laboratory & Field Skills:** Soil testing, building material testing, RTK survey & site audit.
- **Professional Skills:** Technical writing, presentation, poster making, leadership.

Achievements

- Research abstract accepted at the [ICPCCE Conference – IIT Bhubaneswar](#).
- Runner-up – Poster Presentation in Civil Engineering ([LinkedIn](#)).
- Winner – Technical Event “Trust a Truss” ([YouTube](#)).
- Poster Presentation – SRM University AP (ESE 2025) ([LinkedIn](#)).

Hobbies & Interests

Hiking, playing sports (badminton, volleyball, basketball), and spending time in calm environments.